

Swathy Ramakrishnan

Data Scientist. ML Engineer

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Experience

Data Scientist Intern

BalanX Bio

Feb 2026 – Present

US (Remote)

- Deployed a RAG-based conversational AI model to production, enabling semantic search and document Q&A over proprietary biomedical data.
- Built and maintained a CI/CD pipeline (GitHub Actions) automating testing, Docker image builds, and deployment to AWS.
- Containerised the full ML stack using Docker, ensuring reproducible builds across development and production environments.
- Integrated Pinecone as the vector database for high-performance embedding storage and nearest-neighbour retrieval at scale.

Data Scientist

Omdena

May 2024 – Oct 2024

New York, USA (Remote)

- Built end-to-end ML pipeline for patient outcome prediction, increasing recall by 77% and reducing missed diagnoses through improved early-intervention models.
- Implemented brain tumour MRI classification using transfer learning with a fine-tuned CNN (PyTorch), achieving **90% accuracy**.
- Designed and deployed RAG pipelines using HuggingFace Transformers and OpenAI API for entity recognition, text classification, and document Q&A.
- Developed a Streamlit web application integrating real-time ML predictions with model monitoring and performance tracking workflows.
- Conducted EDA and created data visualisations (Matplotlib, Seaborn) to communicate key insights to non-technical stakeholders.
- Collaborated with cross-functional international teams; mentored peers on ML best practices, code review, and deployment workflows (Agile/Scrum).

Data Scientist

Tata Consultancy Services

Jan 2019 – May 2021

Trivandrum, India

- Automated manual model prediction workflows using Python (Pandas, NumPy, Scikit-learn), eliminating repetitive hand-run scoring and freeing analyst time for higher-value tasks.
- Built supervised and unsupervised ML models (regression, classification, clustering) on customer datasets; findings packaged into client deliverables to support data-driven decision making.
- Enhanced team productivity by mentoring junior members and providing technical guidance on Python-based analytical workflows.
- Led Scrum ceremonies and maintained on-time milestone delivery across multiple client projects.

Data Scientist Trainee

Activeologica Lifescience Innovations

Mar 2018 – Jan 2019

Kochi, India

- Increased operational efficiency by 15% by deploying calculation-capable chatbots on Microsoft Azure using trained NLP models.
- Gained hands-on exposure to the Hadoop ecosystem (HDFS) during data pipeline configuration and testing, supporting big data infrastructure in production workflows.

Projects

Multi-Language AI Agent Agno framework, LLM routing, Streamlit, Python [\[demo\]](#)

- Built a multi-language AI assistant that auto-detects user language (German/English) and routes conversations to specialised language agents using the Agno framework; deployed as a live Streamlit app.

URL Summariser – RAG Pipeline LangChain, ChromaDB, OpenAI API, Streamlit [\[demo\]](#)

- Designed a **RAG**-based tool that ingests multiple URLs, stores embeddings in **ChromaDB** (vector database), and answers natural-language questions over the retrieved content.

LLM-Based Book Chatbot LLM, CSV data ingestion, Streamlit [\[demo\]](#)

- Conversational AI that answers book-related questions by combining structured CSV data with LLM reasoning, enabling semantic search over a large book catalogue.

Customer Churn Prediction Python, Scikit-learn, XGBoost, Pandas, Streamlit [\[demo\]](#)

- End-to-end churn prediction pipeline: feature engineering, model training (XGBoost), cross-validation, and a Streamlit dashboard for business users.

Garbage Classification – ResNet PyTorch, Transfer Learning, CNN, Streamlit [\[demo\]](#)

- Fine-tuned a **ResNet** model for image-based garbage classification, achieving **94% accuracy**; deployed as an interactive Streamlit app.

Google Gemini Chatbot Google Gemini API, Python, Streamlit [\[demo\]](#)

- General-purpose conversational AI leveraging the **Google Gemini API** for context-aware, multi-turn dialogue with live deployment.

Technical Skills

Languages: Python (Pandas, NumPy, Scikit-learn), SQL

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM

Generative AI & LLMs: LangChain, HuggingFace Transformers, OpenAI API, Google Gemini, Llama, Claude, RAG, Prompt Engineering, Fine-tuning (LoRA/PEFT), Agno framework, CrewAI

Vector Databases: ChromaDB, FAISS, Pinecone

NLP: Named Entity Recognition, Text Classification, NLTK, SpaCy

Data & Pipelines: Spark (PySpark), Hadoop, MySQL, MongoDB, ETL pipelines

Cloud Platforms: AWS (SageMaker), Microsoft Azure, GCP

Deployment & MLOps: Streamlit, FastAPI, Docker, CI/CD (GitHub Actions), Git, Jupyter, VS Code

Statistics: A/B Testing, Hypothesis Testing, Regression Analysis, Bayesian Methods

Visualisation: Matplotlib, Seaborn, Power BI, Plotly

Methodology: Agile / Scrum, Cross-functional collaboration, Model monitoring

Education

M.Tech – Information Technology

2015 – 2017

Rajagiri School of Engineering and Technology

Thesis: Incremental Model for Analysing Mobile Reviews

B.Tech – Information Technology

2011 – 2015

Rajagiri School of Engineering and Technology

Thesis: Filtering of Online Social Network User Walls

Languages

English – Fluent | **German** – Business fluent